

Vivek Balasaheb Sabale

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Education

- 2021 – 202X  **Ph.D. Quantum Information and Computation, Indian Institute of Technology, Jodhpur.**
Thesis title: *Dynamics of multiqubit nonlocal correlations*
- 2019 – 2021  **M.Sc. Chemistry, Indian Institute of Technology, Madras.**
Thesis title: *Entangled quantum states and correlations in quantum mechanics* .
- 2016 – 2019  **B.Sc. Chemistry, Fergusson College, Pune.**

Research Publications

Journal Articles

- 1 S. Bhattacharya, V. B. Sabale, and A. Kumar, “Heisenberg spin networks for realizing quantum battery with the aid of dzyaloshinskii–moriya interaction,” *New Journal of Physics*, vol. 28, no. 1, p. 014 508, 2026.
- 2 D. Sharma, V. B. Sabale, P. Singh, and A. Kumar, “Harnessing quantum support vector machines for cross-domain classification of quantum states,” *Quantum Machine Intelligence*, vol. 7, no. 1, p. 49, 2025.
 DOI: <https://doi.org/10.1007/s42484-025-00274-4>.
- 3 A. Abu-Nada, S. Banerjee, and V. B. Sabale, “Exploring the non-markovian dynamics in depolarizing maps,” *Phys. Rev. A*, vol. 110, p. 052 209, 5 Nov. 2024.  DOI: 10.1103/PhysRevA.110.052209.
- 4 V. B. Sabale, N. R. Dash, A. Kumar, and S. Banerjee, “Facets of correlated non-markovian channels,” *Annalen der Physik*, vol. 536, no. 10, p. 2 400 151, 2024.  DOI:
<https://doi.org/10.1002/andp.202400151>.
- 5 V. B. Sabale, A. Kumar, and S. Banerjee, “Toward realization of universal quantum teleportation using weak measurements,” *Annalen der Physik*, vol. 536, p. 2 300 392, 4 Apr. 2024, ISSN: 1521-3889.  DOI:
10.1002/ANDP.202300392.

Skills

- Languages  Strong reading, writing and speaking competencies in English, Hindi and Marathi.
- Coding  MATLAB, Mathematica, Python, Qiskit, Julia

Miscellaneous Experience

Awards and Achievements

- 2024  **International Travel Grant**, received financial assistance from Anusandhan National Research Foundation (ANRF) for participating in "Quantum Techniques in Machine Learning, University of Melbourne (25 November 2024 to 29 November 2024)".
-  **Workshop with QIndia**, teaching quantum computing at workshop organised in IIIT Dharwad.
- 2018  **Mahatma Jyotirav Phule Guvant Vidyarthi Scholarship**, Pune University.

Miscellaneous Experience (continued)

Certification

- 2021  **Certificate**, for completing IBM Quantum Challenge Awarded by IBM.
- 2019  **Diploma Certificate**, Project Orientated Chemistry Education (POCE) program of Jawahar Nehru Centre for Advanced Scientific Research, Bangalore.

References

Prof Atul Kumar

Professor

IIT Jodhpur,

<https://atulk4.wixsite.com/atulk>

Prof Subhashish Banerjee

Professor

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